Global–Local Residual Fusion Network for Hyperspectral Image Classification

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***Abstract*—As hyperspectral images (HSIs) continue to increase in data resolution and information richness, current deep learning models need to enhance their feature extraction and understanding capabilities for classification tasks. The complementarity between convolution and attention mechanisms in deep learning enables the capture of both local and global information. However, it faces the problems of the intrinsic coupling of different operators and the precise fusion of different features. In this study, a novel global–local residual fusion network (GLRFNet) is proposed to improve the HSI classification. First, a feature projection with multiple kernels is designed to generate the feature pool before deep extraction and enhance the information connection between operators. Then, a global–local residual (GLR) feature extraction network is built to capture both fine-grained details and large-scale dependencies, improving the feature perception in various classification scenes. It consists of local convolution, global attention, and residual construction branches in a separate and coupled manner. Finally, an optimized inverted bottleneck fusion (IBF) module is built to perform nonlinear and comprehensive feature fusion for mining the high-level semantics and understanding the category relationships. The experiments on four HSI datasets demonstrate the superiority of GLRFNet compared to other state-of-the-art methods, especially its classification performance under small sample conditions, with higher recognition accuracy and better class boundaries. In addition, parameter analysis and ablation experiments are also conducted to determine the optimal parameters and verify the module effectiveness.**

***Index Terms*—Global–local feature extraction, hyperspectral image (HSI) classification, multikernel feature projection, nonlinear feature fusion, residual structure.**

I. Introduction

HYPERSPECTRAL image (HSI) is a crucial type of image data that combines the spatial information with high-resolution spectral information. It captures the spectrum in the range from visible to near-infrared wavelength, allowing for the detailed representation of materials and objects on the ground [1]. Due to these unique characteristics, HSIs have found extensive applications in agriculture, forestry, biodiversity, mineralogy, hydrology, environment, and urban planning [2], [3], [4]. Currently, there are three main platforms for acquiring HSIs: aerospace platforms such as the GaoFen-5 satellite and the Earth Observing-1 satellite, airborne platforms, and unmanned aerial vehicles (UAVs) [5]. Among these platforms, UAVs are particularly suitable for studying land-cover classification in urban due to their operational flexibility, lower cost, and flight capability [6].

Among the various applications of HSIs, the classification task has obtained the most attention from researchers, which involves determining the land-cover type associated with each pixel [7]. In the early stages of HSI classification research, classical classification methods were proposed and utilized to analyze the spectral features present in HSIs, including support vector machine (SVM) [8], multinomial logistic regression [9], and dynamic random subspace approach [10]. Similarly, other methods explored the role of spatial features in HSI classification [11], which led to the development of a spectral–spatial feature fusion and classification method. For instance, Camps-Valls et al. [12] utilized composite kernels to fully extract the joint spectral–spatial information in HSIs, and Fauvel et al. [13] employed spatial–spectral kernels for the same purpose. However, traditional machine learning methods require a significant amount of prior knowledge and feature design, and lack the ability for deep extraction and adaptive recognition in complex scenes. Recently, with the continuous advancement of deep learning [14], [15], it has gained widespread popularity across various fields, including HSI classification [16], [17].

In the field of HSI classification, the convolutional neural network (CNN) and Transformer model [18] in deep learning have demonstrated impressive performance. The 3-D U-Net Transformer model proposed by Jamali et al. [19] dynamically synthesizes the wetland HSI samples during training based on the availability of each class. Gibril et al. [20] evaluated the performance and generalization ability of various algorithms for classifying palms in multiscale HSIs from UAV. He et al. [21] proposed a transformer-based bidirectional encoder representation method, which has global receptive field to capture the dependencies between pixels. As for CNN models in HSI classification, Yang et al. [22] utilized a 2-D CNN to extract joint spectral–spatial features from HSIs for classification.

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Lee and Kwon [23] and Paoletti et al. [24] introduced residual learning to construct very deep and wide CNNs for extracting more discriminative features in HSIs. Teng et al. [25] combined object-oriented CNN and deep transfer learning to classify tree species in high-resolution HSIs of forest parks.

To mitigate the limitations imposed by scarce labeled samples, researchers have developed unsupervised deep learning approaches to obtain stable HSI features. Lin et al. [26] proposed a spatial–spectral masked autoencoder (SS-MAE) for joint classification of HSI and other data. Cao et al. [27] achieved performance improvements through the integration of contrastive loss with Transformer-based masked autoencoders. Further advancing this direction, Zhou et al. [28] developed a framework that combines multiscale masked Transformers with local contrastive learning. Wang et al. [29] contributed the nearest-neighbor contrastive learning network (NNCNet), which integrates unsupervised learning with attention mechanisms. Parallel to these developments, significant attention has been directed toward multiscale feature extraction strategies. Huang et al. [30] pioneered a superpixel-based framework that combines multi-instance factor analysis with multiscale decision fusion. He et al. [31] employed multiscale self-attention mechanisms and dynamic feature aggregation for HSI analysis. Complementing these approaches, Ding et al. [32] introduced a sophisticated architecture that integrates superpixel segmentation, spectral feature extraction, and multiscale graph feature fusion, enhancing both classification accuracy and noise robustness.

Previous works based on convolution or attention mechanisms have been effective in completing the tasks of HSI classification, but the shortcomings of a single operator in model performance become evident as the complexity and diversity of classification increase. Consequently, multioperator methods were developed in HSI classification, proposing models that can integrate the advantages of different operators to overcome the limitations [33], [34]. For example, dual-attention transformer encoder (DATE) [35] and multiattention Transformer [36] were built to combine multiple attention operators, and cross-mixed residual network [37] was established to integrate multiple convolution operators. The current research landscape reveals a growing trend toward hybrid architectures of CNNs and Transformers for HSI classification [38], [39]. Notably, Ouyang et al. [40] accomplished this paradigm by augmenting Transformer-based models with CNN operators to enhance local representation. Similarly, Xu et al. [41] proposed a multiscale spectral and spatial feature extraction framework with a dual-branch hybrid architecture. Further refining this approach, Zhou et al. [42] optimized the operator integration using a spatial–spectral fusion projection and an interactive self-attention module. Moreover, Xu et al. [38] proposed the cross-attention fusion network (CAFFormer), featuring a dynamic-CNN branch for local extraction and a Gaussian Transformer branch for global modeling.

However, although some studies have recognized the complementary attributes of convolution and attention mechanisms, these methods still treat them as separate entities rather than integrated systems. First, they perform separate feature extraction and then combine the extracted features [41], or treat one as an extension module or replacement block of the other [40]. This architectural segregation results in inadequate intrinsic coupling between operators, limiting their collective capacity to capture fine-grained local attributes and long-range global dependencies. Consequently, these models might fail to maintain the fundamental and stable characteristics of multiple operators. Second, some existing methods fuse convolution and Transformer in both serial and parallel mechanisms [39]. Nevertheless, due to the lack of deep multistage local and global feature interactions, they cannot effectively represent sophisticated object features and subtle spectral–spatial variations, especially in challenging scenarios with high intraclass variability and interclass similarity. Third, multioperator feature fusion methods using interactive self-attention [43] and cross-attention [38] have been proposed in existing studies, but these methods also incur notable computational overhead. It is important to accurately consider the contribution of different operators and the complexity of nonlinear relationships in the feature fusion stage for HSI classification.

To address the aforementioned problems, a novel global–local residual fusion network (GLRFNet) is proposed for effective HSI classification, integrating different operators, as shown in Fig. 1. It mainly contains the multikernel feature projection, global–local residual (GLR) feature extraction, and optimized inverted bottleneck fusion (IBF). First, the framework initiates with a multiscale feature projection module employing parallel convolutional kernels of dimensions 1×1, 3×3, and 5×5, which generates a shared feature pool encompassing shallow representations. By organizing projection information across multiple kernels, the module establishes a semantically consistent foundation for subsequent feature extraction in diverse operator branches. This design significantly enhances both the intrinsic coupling and cross-operator stability of extracted features. Then, the core feature extraction module integrates convolutional operators with attention mechanisms through an innovative tribranch architecture. GLR-G branch specializes in capturing long-range global dependencies through self-attention mechanisms, GLR-L branch focuses on extracting fine-grained local attributes using optimized convolutional operations, and GLR-R branch constructs hierarchical feature representations through spectral–spatial residual learning. The synergetic combination of these branches further deepens the multioperator feature interaction through multiple-stage stacking. Finally, the optimized IBF module performs advanced fusion of the heterogeneous features generated by the GLR module, obtaining robust and enriched representations that comprehensively encode ground object information. Compared to conventional fusion approaches, IBF introduces multilevel nonlinear transformation along the channel dimension, adaptive feature recalibration, and efficient cross-branch interaction. The hierarchical organization from projection to extraction to fusion creates a coherent pipeline for HSI feature learning.

The contributions of GLRFNet can be summarized as follows.

1) In order to enhance the intrinsic connection between different operators, a feature projection process utilizing

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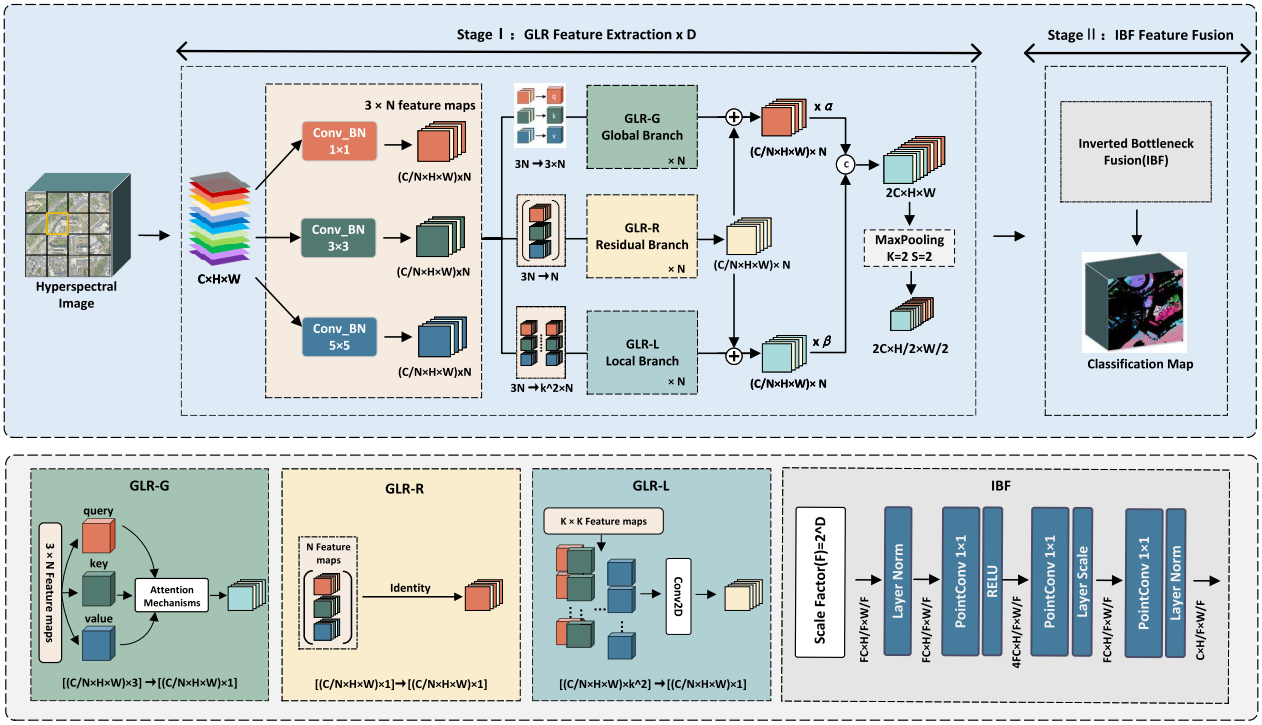


Fig. 1. Flowchart of the proposed GLRFNet for HSI classification.

multiple kernels is designed before feature extraction. The multikernel projection enables diverse contextual perception from the input image, providing a comprehensive foundation of information organization for the subsequent local and global feature extraction. The shared feature pool strengthens the internal coupling and feature stability between convolution and attention operators, which maintains the semantic consistency.

2) A novel GLR feature extraction network is proposed to capture the local and global information of HSI in a residual construction manner. The convolution operator is essential for detecting edges, textures, and other local attributes, and attention operator enables the model to capture long-range dependencies and understand the global context. Furthermore, GLR takes the advantage of residual structure to integrate diverse spectral and spatial features, which provides an effective approach to handle different operators for HSI classification.
3) An optimized IBF module is built to further fuse the operator content and mine the classification semantics in a nonlinear manner. IBF enhances the traditional inverted bottleneck by deepening the fusion layers and increasing the fitting ability, to capture more intricate relationships and more complicated structures. The nonlinear fusion enables the network to integrate diverse semantics from different channels, thus improving the overall representation of output features and understanding the category mapping relationships.

II. Related Works

A. Attention Mechanism

The attention mechanism, inspired by the biological process of human vision, has become a popular model in various computer vision (CV) tasks [18]. It simulates the brain's ability to quickly scan an entire image and identify the areas of interest. By allocating more attention resources to these specific areas, the attention mechanism extracts crucial features that are significant for the task. In essence, the attention mechanism aims to extract the most relevant information from a vast amount of data, enabling more effective and focused analysis. The self-attention mechanism computes attention weights for each position by comparing the similarity between the query, key, and value vectors. The self-attention weights determine how much each position contributes to the representation of the current position.

Regarding the application, an attention-based module with N heads can be described as follows. The input feature is denoted as $F \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, and the output feature is denoted as $G \in \mathbb{R}^{C_{\text{out}} \times H \times W}$. Here, C_{in} and C_{out} represent the channel number of input and output features, while H and W represent the height and width dimensions, respectively. To describe the attention calculation, let $F_{ij} \in \mathbb{R}^{C_{\text{in}}}$ and $G_{ij} \in \mathbb{R}^{C_{\text{out}}}$ denote the input and output tensors corresponding to the pixel at position (i, j). Additionally, $\mathcal{N}_k(i, j)$ denotes the spatial local area centered around (i, j) with a range of k. The attention calculation can be expressed as

$$G_{ij} = \|_{l=1}^{N} \left(\sum_{a,b \in \mathcal{N}_k(i,j)} \mathcal{A}\left(W_q F_{ij}, W_k F_{ab}\right) W_v F_{ab} \right) \tag{1}$$

where $\|$ represents the output concatenation of N heads. W_q, W_k, and W_v are the projection matrices of queries, keys, and values in the attention mechanism, respectively. $\mathcal{A}(W_q F_{ij}, W_k F_{ab})$ is the calculation process of attention

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weights in the $\mathcal{N}_k(i,j)$ area, which can be expressed as

$$\mathcal{A}\left(W_qF_{ij}, W_kF_{ab}\right) = \underset{\mathcal{N}_k(i,j)}{\text{Softmax}}\left(\frac{\left(W_qF_{ij}\right)^T\left(W_kF_{ab}\right)}{\sqrt{d}}\right) \tag{2}$$

where d is the feature dimension of W_qF_{ij}.

B. Convolution Mechanism

Convolution is a powerful deep learning model that has been widely used for processing multidimensional data, such as images and sounds [44]. The concept of convolution is inspired by the biological structure of visual cortex [45]. In the convolution, a convolution kernel slides over the input data in a systematic manner, such as an image. At each position, the kernel is multiplied with the corresponding input content within the sliding window. These products are, then, integrated or summed up after sliding the kernel across the entire input. This process reflects the similarity or correlation between input data and convolution kernel. CNN is a type of neural network that utilizes convolutional layers. CNNs automatically learn and extract relevant features in the hidden layers during the training process of the classifier [46].

To describe the convolution operations, the convolution kernel is represented by $K \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times k\times k}$. Here, k is the kernel size, and C_{in} and C_{out} are the input and output channel sizes, respectively. Consider the input and output feature maps represented by tensors $F \in \mathbb{R}^{C_{\text{in}}\times H\times W}$ and $G \in \mathbb{R}^{C_{\text{out}}\times H\times W}$. The feature tensors $F_{ij} \in \mathbb{R}^{C_{\text{in}}}$ and $G_{ij} \in \mathbb{R}^{C_{\text{out}}}$ correspond to the values of pixel (i,j) in F and G, respectively. Then, the convolution can be expressed as

$$G_{ij} = \sum_{p,q} K_{p,q}F_{i+p-\lfloor k/2\rfloor, j+q-\lfloor k/2\rfloor} \tag{3}$$

where $K_{p,q} \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$, and $p,q \in \{0,1,\ldots,k-1\}$ represents the kernel weight of position (p,q). Further, this formula can be rewritten for convenience as the sum of feature maps with kernels at different positions, which is expressed by

$$G_{ij} = \sum_{p,q} G_{ij}^{(p,q)} \tag{4}$$

$$G_{ij}^{(p,q)} = K_{p,q}F_{i+p-\lfloor k/2\rfloor, j+q-\lfloor k/2\rfloor} \tag{5}$$

where $G_{ij}^{(p,q)}$ represents the feature map of a single kernel.

The shift operation $\tilde{F} = \text{Shift}(F, \Delta x, \Delta y)$ in convolution can be defined as

$$\tilde{F}_{i,j} = F_{i+\Delta x, j+\Delta y} \tag{6}$$

where Δx and Δy represent the horizontal and vertical displacements, respectively. Then, the feature map formula of a single kernel can be rewritten as

$$G_{ij}^{(p,q)} = \text{Shift}\left(K_{p,q}F_{ij}, p-\lfloor k/2\rfloor, q-\lfloor k/2\rfloor\right). \tag{7}$$

Let RF_l and RF_{l+1} denote the size of receptive field at previous and current layers, respectively. S_k and S_s represent the size of the convolution kernel and the convolution step at this layer, respectively. Then, the size of the receptive field obtained by an element in the original input layer can be expressed as

$$RF_{l+1} = RF_l + (S_k - 1)\times S_s. \tag{8}$$

It is evident that the size of the receptive field is directly influenced by the kernel size and the stride of the convolution kernel. However, using a large stride in convolution can result in excessive information loss in the feature map. Hence, a more effective approach to increase the receptive field is to increase the size of the convolution kernel.

III. GLRFNet

This section provides a detailed explanation of the proposed GLRFNet and its key components, including the multikernel feature projection, GLR feature extraction, and optimized IBF, as shown in Fig. 1. Concerning the overall process of GLRFNet, it initiates by performing feature projection on the input HSI with various receptive fields, generating a feature pool consisting of diverse shallow features. Next, different operators are employed to extract distinct but related global and local features from the feature pool simultaneously. Finally, the features extracted by different operators are effectively fused to incorporate sufficient nonlinear variation.

A. Feature Projection With Multiple Kernels

At the beginning of GLRFNet, a multikernel feature projection process is designed to enhance intrinsic coupling between different operators and provide diverse contextual representations. Specifically, the input HSI undergoes parallel convolution operations with kernel sizes of 1×1, 3×3, and 5×5, each followed by batch normalization (BN). This process transforms the input data with size $C\times H\times W$ into three sets of feature maps, each sized $(C/N\times H\times W)\times N$, with varying receptive fields. C, H, and W mean the channel, height, and width dimensions, respectively, and N represents dividing the channels into N groups. These feature maps are, then, integrated into a unified shallow feature pool, which enriches the diversity of feature extraction perspectives. At the same time, it improves semantic consistency and strengthens the coupling between convolution and attention mechanisms. Compared to directly inputting the original HSI into neural operators, this method preorganizes the input data into structured shallow features, providing a robust and comprehensive foundation for subsequent local and global feature extraction. By leveraging this multikernel design, GLRFNet achieves more stable feature representation and improved operator interaction, leading to more robust classification.

Let $\boldsymbol{\mu}^{(k)}, \boldsymbol{\sigma}^{(k)}, \boldsymbol{\gamma}^{(k)}$, and $\boldsymbol{\beta}^{(k)}$ denote the accumulated mean, standard deviation, scaling factor, and bias of BN layer following the convolution with kernel size k, respectively. $C^{(k)}$ denotes the convolution kernel, and * denotes the convolution operator. For input patch $M^{(\text{in})} \in \mathbb{R}^{N\times(C/N)\times H\times W}$, the multikernel feature projection can be expressed as

$$M^{(\text{out},1)} = \text{BN}\left(M^{(\text{in})} * C^{(1)}, \boldsymbol{\mu}^{(1)}, \boldsymbol{\sigma}^{(1)}, \boldsymbol{\gamma}^{(1)}, \boldsymbol{\beta}^{(1)}\right) \tag{9}$$

$$M^{(\text{out},3)} = \text{BN}\left(M^{(\text{in})} * C^{(3)}, \boldsymbol{\mu}^{(3)}, \boldsymbol{\sigma}^{(3)}, \boldsymbol{\gamma}^{(3)}, \boldsymbol{\beta}^{(3)}\right) \tag{10}$$

$$M^{(\text{out},5)} = \text{BN}\left(M^{(\text{in})} * C^{(5)}, \boldsymbol{\mu}^{(5)}, \boldsymbol{\sigma}^{(5)}, \boldsymbol{\gamma}^{(5)}, \boldsymbol{\beta}^{(5)}\right) \tag{11}$$

where the outputs after applying convolutions with kernel sizes $k \in \{1,3,5\}$ are denoted as $M^{(\text{out},k)} \in \mathbb{R}^{N\times(C/N)\times H\times W}$. BN$(\cdot)$

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represents the batch normalization operation, which can be expressed as follows:

$$\begin{aligned}&\mathrm{BN}\left(M^{(\mathrm{in})} * C^{(k)}, \boldsymbol{\mu}^{(k)}, \boldsymbol{\sigma}^{(k)}, \boldsymbol{\gamma}^{(k)}, \boldsymbol{\beta}^{(k)}\right)_{:,i,:,:} \\ &= \left(\left(M^{(\mathrm{in})} * C^{(k)}\right)_{:,i,:,:} - \boldsymbol{\mu}_i^{(k)}\right) \times \frac{\boldsymbol{\gamma}_i^{(k)}}{\boldsymbol{\sigma}_i^{(k)}} + \boldsymbol{\beta}_i^{(k)} \end{aligned} \tag{12}$$

where $1 \le i \le C/N$. After obtaining the feature pool with multikernel shallow features, GLRFNet further generates different variations to provide input for GLR-G, GLR-L, and GLR-R branches, respectively. This approach aims to fully explore the abundant spatial–spectral features in HSI at different levels and biases for different operators.

Specifically, GLR-G branch utilizes the feature understanding capability of attention mechanism to extract global features of ground objects in HSI. The previously extracted features with varying receptive fields are used to map and acquire queries, keys, and values, in order to raise the diversity. First, the features obtained through 1×1 convolution exhibit linear transformation along channel dimension with respect to the input, and GLR-G selects these features with minimal divergence from the input as keys. Second, following the 3×3 convolution, the extracted features encompass some shallow information within a small context, and GLR-G employs these features as queries to participate in the similarity calculation with keys. Third, the features derived through 5×5 convolution possess a larger receptive field with more contextual information, and GLR-G regards these features as values to provide the foundation for attention calculation. Then, the GLR-G branch can be described as

$$k_{ij} = W_k \times M_{ij}^{(\mathrm{out},1)} \tag{13}$$

$$q_{ij} = W_q \times M_{ij}^{(\mathrm{out},3)} \tag{14}$$

$$v_{ij} = W_v \times M_{ij}^{(\mathrm{out},5)} \tag{15}$$

where $M_{ij}^{(\mathrm{out},k)}$ denotes the feature at spatial location (i, j) from the output of convolution with kernel size $k \in \{1, 3, 5\}$. W_k, W_q, and W_v are the learnable matrices that map the feature to key k_{ij}, query q_{ij}, and value v_{ij}, respectively.

The GLR-L branch employs the feature mining capability of the convolution mechanism and focuses on capturing local detailed information. After combining the shift and aggregation of convolution, it is decided to concatenate all feature maps in the feature pool to obtain the input of GLR-L branch. The concatenation of multikernel shallow features is beneficial to enhancing the comprehensive foundation for subsequent deep local feature extraction. This process can be represented as

$$FP_{ij} = \left[M_{ij}^{(\mathrm{out},1)}, M_{ij}^{(\mathrm{out},3)}, M_{ij}^{(\mathrm{out},5)}\right] \tag{16}$$

where FP_{ij} represents the concatenated result of feature maps.

The GLR-R branch assists the GLR-G and GLR-L branches by constructing a residual structure. This structure addresses the degradation problem of the network model and the issue of gradient vanishing. The residual structure facilitates the GLR-G to focus on extracting global connections, as well as the GLR-L to focus on extracting detailed information, which directly transmits the shallow features. It ensures the effectiveness of model training and the performance on classification tasks. With this design, all the feature maps are collected and compressed to a size of $(H \times W \times C/N) \times N$, which serves as the input for the GLR-R branch.

B. GLR Feature Extraction

Compared to previous methods that rely on single operators or directly combine different operators, the features obtained by GLRFNet exhibit superior capability in addressing complex tasks. This advantage stems from the GLR feature extraction module, which consists of three interacting branches: the GLR-G, GLR-L, and GLR-R. On the one hand, the global and local features share a common feature pool foundation designed with multikernel information projection, which strengthens the intrinsic coupling and enhances the multioperator feature stability. On the other hand, the attention and convolution mechanisms are effectively integrated through the residual structure and nonlinear fusion, ensuring robust and stable feature extraction. Each branch focuses on a specific aspect of feature extraction and interacts with the others to form a complementary and cohesive representation. The following is a detailed description of the roles and interactions of GLR-G, GLR-L, and GLR-R branches.

1) GLR-G: The GLR-G branch leverages an attention mechanism to capture long-range dependencies and establish global context relationships. By prioritizing feature extraction from important areas and integrating features across different regions, it provides a high-level abstraction of the scene, offering a global perspective for semantic understanding. The multikernel shallow features in the feature pool enhance the attention computation of GLR-G, resulting in robust global dependencies. Specifically, the keys, queries, and values are mapped from the information of 1×1, 3×3, and 5×5 convolution kernels, respectively.

After mapping the keys, queries, and values that are represented by k_{ij}, q_{ij}, and v_{ij}, the GLR-G calculation process can be represented as

$$G_{ij} = \|_{l=1}^{N} \left(\sum_{a,b \in \mathcal{N}_k(i,j)} \mathrm{A}\left(q_{ij}, k_{ab}\right) \times v_{ab} \right) \tag{17}$$

where $\|$ represents the concatenation of the outputs from N group of GLR-G operations. Within the region $\mathcal{N}_k(i, j)$, $\mathrm{A}(q_{ij}, k_{ab})$ denotes the process of obtaining the update matrix by calculating the similarity using queries and keys. $\mathrm{A}(q_{ij}, k_{ab}) \times v_{ab}$ represents the update values obtained by weighting values with the matrix. G_{ij} represents the GLR-G output obtained from the corresponding input patch centered at ith row and jth column.

2) GLR-L: The GLR-L branch leverages convolution operators to mine fine-grained details and capture local structures. By aggregating multiscale shallow features from the feature pool, GLR-L preserves spatial information that is critical for distinguishing subtle differences between classes. To achieve this, GLR-L first adjusts the number of feature maps in the feature pool using fully connected layers, enabling subsequent shift and aggregation operations. The adjusted feature maps are, then, sequentially projected to different positions (p, q), where $p, q \in \{0, 1, \ldots, k-1\}$ and k represents the convolution

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kernel size. These projections undergo shift and aggregation processes, allowing GLR-L to effectively extract diverse and detailed local features.

After obtaining FP_{ij} with a size of $(C/N \times H \times W) \times 3N$, it is expanded through fully connected operations into a set of N group feature maps l_{ij} with a size of $(C/N \times H \times W) \times k^2$. For each l_{ij}, GLR-L first generates k^2 feature maps and then performs linear projections on them based on different kernel weights $K_{p,q}$, resulting in the tensor $l_{ij}^{(p,q)}$ for each location (p, q), where $p, q \in \{0, 1, \ldots, k-1\}$. This process can be represented as

$$l_{ij}^{(p,q)} = K_{p,q} l_{ij}. \tag{18}$$

Then, GLR-L shifts $l_{ij}^{(p,q)}$ according to k, expressed as

$$L_{ij}^{(p,q)} = \text{Shift}\left(l_{ij}^{(p,q)}, p - \lfloor k/2 \rfloor, q - \lfloor k/2 \rfloor\right) \tag{19}$$

where the Shift$(\cdot)$ is implemented as $\text{Shift}(F, \Delta x, \Delta y) = F_{i+\Delta x, j+\Delta y}, \forall i, j$. Following that, GLR-L aggregates the k^2 obtained $L_{ij}^{(p,q)}$ to complete the calculation process of each I_{ij} in GLR-L. It, then, obtains a set of local features $\sum_{p,q} L_{ij}^{(p,q)}$ corresponding to the input patch. Finally, GLR-L concatenates the N groups of local features as the output L_{ij} through

$$L_{ij} = \|_{l=1}^{N} \left(\sum_{p,q} L_{ij}^{(p,q)}\right). \tag{20}$$

3) GLR-R: Due to the complexity of the network, GLRFNet encounters a bottleneck during the training process. This is primarily because it extracts a large number of features for global and local targets, making it difficult to effectively integrate these diverse features. Moreover, GLRFNet faces issues such as network degradation and gradient vanishing. To address these challenges, GLRFNet incorporates the GLR-R branch to implement a residual mapping. Acting as a bridge, the residual branch transmits shallow features directly to the output, ensuring effective gradient flow and mitigating issues like vanishing gradients. Additionally, GLR-R enhances the interaction between GLR-G and GLR-L by preserving shared feature semantics and maintaining consistency across branches, without introducing more parameters.

Specifically, the GLR-R branch first collocates all the feature maps in the feature pool. It compresses the collocated feature map set from size $(C/N \times H \times W) \times 3N$ to size $(C/N \times H \times W) \times N$ using a 1×1 convolution. Then, the compressed feature map set is taken as the GLR-R output R_{ij}. It is added with G_{ij} and L_{ij} to obtain the residual features GR_{ij} and LR_{ij}, respectively. Additionally, learnable scalars α and β are utilized to weight the two residual features. The weighted residual features are concatenated in the channel dimension to produce features GLR_{ij} as the output of GLR. This process can be expressed as

$$\begin{aligned} \text{GLR}_{ij} &= \left[\alpha\left(G_{ij} + R_{ij}\right), \beta\left(L_{ij} + R_{ij}\right)\right] \\ &= \left[\alpha\left(GR_{ij}\right), \beta\left(LR_{ij}\right)\right]. \end{aligned} \tag{21}$$

In summary, the GLR-G, GLR-L, and GLR-R branches complement each other, which forms a comprehensive feature extraction framework that captures global, local, and residual information, enhancing discrimination of spatial–spectral variations. By sharing a multikernel feature pool, these branches ensure both the unification and diversity of global and local features, thereby strengthening their intrinsic coupling. It is noteworthy that the multikernel feature projection and GLR feature extraction modules are repeated for D iterations. Assuming that at the ith iteration, the input feature dimensions are $2^{i-1}C \times H/2^{i-1} \times W/2^{i-1}$, after these two modules, the output feature dimensions become $2^i C \times H/2^i \times W/2^i$. Thus, with each iteration, the channel number doubles while the spatial dimensions are halved. Through iterative processing, the fused features become more compact and coherent, significantly enhancing the model's feature representation and interaction capability.

C. Optimized IBF

Feature fusion is a widely used operation in modern network architectures, and it can be categorized into two types of early fusion and late fusion, based on the order of fusion and prediction. Early fusion involves merging features extracted through different methods and applying classifiers to the fused features for prediction [47]. Late fusion improves performance by combining the prediction results of classifiers on different features [48]. Considering early fusion is beneficial for exploring complementary relationships between different operators at the feature level, GLRFNet designs a nonlinear IBF module for attention and convolution features. On the one hand, the common bottleneck layer reduces the number of parameters in the computation process for efficient feature extraction. On the other hand, the common inverted bottleneck layer achieves performance improvement by exponentially expanding the dimension of data. Differently, the proposed IBF module optimizes the design of the inverted bottleneck layer by shifting the functional layer downward and keeping the nonlinear operations between upscaling and downscaling convolutions. This is done to achieve the appropriate and sufficient nonlinear variation before fusion. It is important to note that the functional layer in common approaches refers to the depth convolution for feature extraction, whereas in the optimized IBF, it represents the point convolution for feature fusion.

The IBF module takes the concatenated features as input to generate more powerful fused features with semantic representations, through multiple nonlinear operations and channel variations. This enables GLRFNet to produce abstract features for HSI and fit the mapping relationships between features and categories. Let D represent the number of GLR iterations, where $D \in \{0, 1, \ldots, \lfloor \log^2(\text{patchsize}) \rfloor\}$. F represents the scaling factor, which is equal to the decrease factor on H and W as well as the increase factor on C after the entire feature extraction process. It is set to 2^D in this study. The size of output from the GLR process is $FC \times H/F \times W/F$, and the main objective of IBF is to compress GLR_{ij} into fused features with size of $C \times H/F \times W/F$, as the final output O_{ij} of GLRFNet. IBF achieves this compression in the channel dimension using an optimized structure on the bottleneck layer. To enhance the nonlinear mapping capability of O_{ij}, IBF applies rectified linear unit (RELU), layer scaling

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(LS), and layer normalization (LN) operations to GLR_{ij}. The entire process of IBF is illustrated in Fig. 1.

Specifically, when GLR_{ij} enters the IBF, it first undergoes an LN layer to normalize along the feature dimension. The choice of LN over BN is because LN's effect is independent of the batch size, and it normalizes all features of each sample individually. This aligns better with the function of IBF, as opposed to BN, which normalizes each feature within a batch. The implementation is as follows:

$$M_{ln}^{(\mathrm{out})} = \frac{M_{ln}^{(\mathrm{in})} - \mathrm{E}\left(M_{ln}^{(\mathrm{in})}\right)}{\sqrt{\mathrm{Var}\left(M_{ln}^{(\mathrm{in})}\right) + \varepsilon}} \times \omega + \delta \tag{22}$$

where $M_{ln}^{(\mathrm{in})}$ represents the input of LN layer, such as GLR_{ij}, and $M_{ln}^{(\mathrm{out})}$ represents the output. $\mathrm{E}(\cdot)$ and $\mathrm{Var}(\cdot)$ denote the mean and variance of input, respectively. ε is a constant added for numerical stability, ω is a learnable scaling parameter, and δ is a learnable bias parameter. Following LN, the number of channels in the LN-processed GLR_{ij} is quadrupled through the first point convolution. The first point convolution changes the size of the feature from $FC \times H/F \times W/F$ to $4FC \times H/F \times W/F$. This expansion is followed by the RELU activation function, enhancing the nonlinearity of features and mitigating the gradient problem. The implementation is as follows:

$$M_{rl}^{(\mathrm{out})} = \begin{Bmatrix} 0, M_{rl}^{(\mathrm{in})} < 0 \\ M_{rl}^{(\mathrm{in})}, M_{rl}^{(\mathrm{in})} > 0 \end{Bmatrix} \tag{23}$$

where $M_{rl}^{(\mathrm{in})}$ represents the input of RELU layer, and $M_{rl}^{(\mathrm{out})}$ represents the output.

Next, GLR_{ij} passes through the second point convolution to compress the number of channels by a factor of four. The second point convolution changes the size of feature from $4FC \times H/F \times W/F$ to $FC \times H/F \times W/F$. This is followed by an LS operation, which scales the values on each channel using learnable 1-D vectors. The LS operation automatically adjusts the features and improves the training quality of GLRFNet. Finally, GLR_{ij} undergoes the third point convolution for the last dimension change, compressing its size from $FC \times H/F \times W/F$ to $C \times H/F \times W/F$. The compressed feature map is, then, subjected to the second LN and eventually output as O_{ij} after this process. Let $\mathrm{IB}(\cdot)$ represent the first two point convolutions responsible for dimension expansion and compression, as well as the nonlinear operations within them. Let $\mathrm{FF}(\cdot)$ represent the last point convolution responsible for feature fusion. Then, the IBF can be summarized as

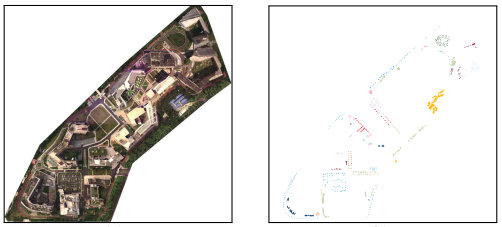
$$\mathrm{GLR}_{ij}^{(\mathrm{IB})} = \mathrm{IB}\left(\mathrm{GLR}_{ij}\right) \tag{24}$$

$$O_{ij} = \mathrm{LN}\left(\mathrm{FF}\left(\mathrm{GLR}_{ij}^{(\mathrm{IB})}\right), \varepsilon, \omega, \delta\right). \tag{25}$$

IV. Experimental Setup

A. Description of HSI Datasets

1) South Yuehai Campus of Shenzhen University: This dataset is captured by a UAV platform in South Yuehai Campus of Shenzhen University (SYCSU), Shenzhen, China. The data have dimensions of 1885×1983 pixels. Out of these, 39 244 pixels are labeled with 14 tree categories. The data consist of 112 bands with a spatial resolution of 0.4 m. Fig. 2 displays the false color image, ground-truth map, and category labels for this dataset. Detailed statistical information for each category is provided in Table I.

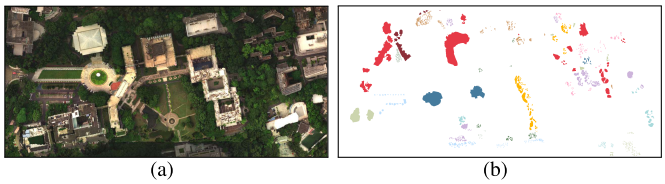


(a) (b)

Fig. 2. (a) False color image and (b) ground-truth map of labels for the SYCSU dataset.

TABLE I
Land-Cover Classes With the Number of Samples per Class for the SYCSU Dataset

Class	Color	Land Cover Type	Number of Samples
C1		Phoenix sylvestris	2,056
C2		Lagerstroemia indica	2,124
C3		Senna siamea	708
C4		Mangifera indica	657
C5		Terminalia mantaly	11,181
C6		Araucaria cunninghamii	1,951
C7		Catalpa	965
C8		Cocos nucifera	1,476
C9		Delonix regia	2,034
C10		Artocarpus nitidus	1,626
C11		Plumeria	2,109
C12		Camphora officinarum	4,866
C13		Leucaena leucocephala	6,391
C14		Bombax ceiba	1,100
		ToTal	39,244



(a) (b)

Fig. 3. (a) False color image and (b) ground-truth map of labels for the NYCSU dataset.

2) North Yuehai Campus of Shenzhen University: This dataset is also captured by a UAV platform in Shenzhen University, with a size of 788×1691 pixels. Out of these, 73 059 pixels are labeled into 11 tree categories. The data contain the same number of bands and spatial resolution as SYCSU, but North Yuehai Campus of Shenzhen University (NYCSU) covers a larger area of vegetation. Fig. 3 displays the false color image, ground-truth map, and category labels for this dataset. Detailed statistical information for each category is provided in Table II.

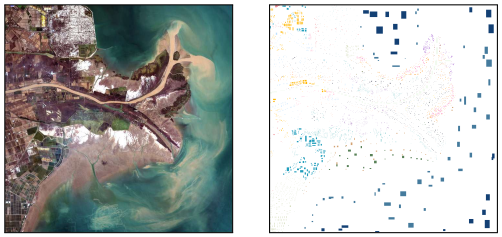
3) Yellow River Estuary: This dataset is captured by China's Gaofen 5 satellite using the advanced hyperspectral imager (AHSI). It consists of 280 bands and has an image size of 1400×1400 pixels. Out of these, 77 937 pixels are labeled into 20 categories. The spatial resolution is 30 m. Fig. 4 illustrates the false color image, ground-truth map, and

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TABLE II

LAND-COVER CLASSES WITH THE NUMBER OF SAMPLES PER CLASS FOR THE NYCSU DATASET

Class	Color	Land Cover Type	Number of Samples
C1		Microcarpa	29,589
C2		Dimocarpus longan	3,915
C3		Catalpa	2,624
C4		Ficus elastica	9,280
C5		Acacia auriculiformis	5,867
C6		Cocos nucifera	1,386
C7		Terminalia arjuna	3,601
C8		Delonix regia	6,235
C9		Ficus virens	1,887
C10		Camphora officinarum	3,042
C11		Swietenia	5,633
		ToTal	73,059



(a) (b)

Fig. 4. (a) False color image and (b) ground-truth map of labels for the YRE dataset.

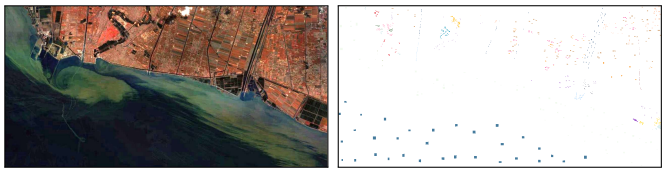
TABLE III

LAND-COVER CLASSES WITH THE NUMBER OF SAMPLES PER CLASS FOR THE YRE DATASET

Class	Color	Land Cover Type	Number of Samples
C1		Building	533
C2		River	5,376
C3		Salt marsh	4,995
C4		Shallow sea	17,550
C5		Deep sea	18,677
C6		Intertidal saltwater marsh	2,343
C7		Tidal flat	1,792
C8		Pond	1,787
C9		Sorghum	646
C10		Corn	1,509
C11		Lotus root	2,719
C12		Aquaculture	8,019
C13		Rice	5,508
C14		Tamarix chinensis	1,220
C15		Freshwater herbaceous marsh	1,417
C16		Suaeda salsa	874
C17		Spartina alterniflora	580
C18		Reed	1,970
C19		Floodplain	347
C20		Locust	75
		Total	77,937

category labels for the Yellow River Estuary (YRE) dataset. Detailed statistical information for each category is provided in Table III.

4) Yan Cheng: This dataset is captured by China's Gaofen 5 satellite in Yancheng City of Jiangsu Province, China. It consists of 266 bands and has an image size of 585×1175 pixels. Out of these, 7112 pixels are labeled into 20 categories. It has a spatial resolution of 30 m. Fig. 5 shows the false color image, ground-truth map, and category labels of the Yan Cheng (YC) dataset. Detailed statistical information for each category is provided in Table IV.



(a) (b)

Fig. 5. (a) False color image and (b) ground-truth map of labels for the YC dataset.

TABLE IV

LAND-COVER CLASSES WITH THE NUMBER OF SAMPLES PER CLASS FOR THE YC DATASET

Class	Color	Land Cover Type	Number of Samples
C1		Sea	2,395
C2		Off shore water	1,305
C3		Salinas	110
C4		Pond	133
C5		Marsh	21
C6		Spartina	83
C7		Mudflate	218
C8		Aquculture	228
C9		Paddy	832
C10		Eatuarine water	172
C11		River	217
C12		Woodland	156
C13		Bare land	154
C14		Architecture	360
C15		Fallow land	234
C16		Dry land	204
C17		Suaeda	85
C18		Mariculyure	35
C19		Ditch	38
C20		Reed	132
		Total	7,112

B. Parameter Setting

Balancing model complexity and efficiency is crucial in developing the classification method, including network depth, division groups, and module iterations. In the experiments of GLRFNet, the number of GLR groups, denoted as N, is set to 8, and the number of GLR iterations, denoted as D, is set to 2. During the training process, the model is trained for 200 epochs using the Adam optimizer with a learning rate of 10^{-3}. For the evaluation, all experimental results are obtained through ten repetitions. The performance of different models in the comparison experiment is evaluated using the overall accuracy (OA), class accuracy (CA), average accuracy (AA), and the Kappa coefficient (Kappa). In the comparison experiments with different numbers of training samples, the range is set to $\{10, 15, 20, \ldots, 50\}$. The number of training samples per class is set to 20 for both the parameter and ablation experiments because the accuracy of each method gradually stabilizes in this case for reliable analysis. On the SYCSU and NYCSU datasets, the tree area is first selected through normalized difference vegetation index (NDVI) calculation and then used for tree species classification.

V. EXPERIMENTAL RESULTS

A. Parameter Analysis for GLR Iteration

In GLRFNet, increasing the number of GLR iterations appropriately can lead to more abstract and expressive feature

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TABLE V
PARAMETER STUDY RESULTS TOWARD DIFFERENT NUMBER OF GLR ITERATIONS ON FOUR DATASETS

Dataset	Indicator	$D = 1$	$D = 2$	$D = 3$	$D = 4$
SYCSU	OA (%)	92.14±1.24	**93.43±1.05**	89.95±1.89	70.25±3.21
	Kappa	0.902±0.013	**0.922±0.011**	0.885±0.017	0.662±0.029
NYCSU	OA (%)	88.36±1.43	**90.87±1.17**	76.00±2.25	61.67±3.45
	Kappa	0.856±0.014	**0.887±0.012**	0.669±0.022	0.442±0.037
YRE	OA (%)	88.24±1.49	**92.14±1.21**	70.90±2.37	41.34±3.96
	Kappa	0.877±0.016	**0.909±0.013**	0.664±0.024	0.326±0.041
YC	OA (%)	94.95±1.08	**96.65±0.93**	80.09±2.06	75.79±2.74
	Kappa	0.940±0.010	**0.960±0.008**	0.767±0.019	0.727±0.026

extraction, which ultimately improves classification performance. However, it is important to consider the variation in feature size when doing so. For instance, if the input size is $C \times H \times W$, a single GLR process will result in an output size of $2C \times H/2 \times W/2$. Repetitions of GLR may cause distortion in the final output's feature sizes, with very small heights and widths but a large number of channels. Taking these factors into account, the number of GLR iterations is tested with $D \in \{1, 2, 3, 4\}$, to determine the most suitable number of GLR iterations for the HSI classification task.

The classification results for different D on four datasets are presented in Table V. It is evident that when $D = 2$, GLRFNet achieves the best results across all four datasets, striking a balance between the quality and size of GLR output features. The OA obtained for the $D = 2$ case is higher than the $D = 1$ case by 1.29%, 2.51%, 3.90%, and 1.70% for the respective datasets. Moreover, compared to the $D = 3$ case, the $D = 2$ case demonstrates an improvement in OA by 3.48%, 14.87%, 21.24%, and 16.56% for the respective datasets. Based on the performance of GLRFNet under different cases, it can be concluded that adding an extra iteration of GLR process results in more robust features for the HSI classification task. However, when performing three iterations of GLR, the quality of extracted features significantly degrades, particularly for the YRE dataset. Ultimately, in the $D = 4$ case, GLRFNet performs remarkably poorly on all four datasets, with an average decay of 31.01% and 0.38 in OA and Kappa, respectively, compared to the $D = 2$ case. It is crucial to maintain a balanced relationship among the channel number (C), height (H), and width (W) of output features. Increasing C allows the GLRFNet to extract spectral information in more detail, while decreasing H and W enables the network to extract spatial features within wider contexts. However, excessive channels and small 2-D dimensions contribute negatively to the classification performance. Instead, they introduce numerous invalid and difficult-to-classify spectral features and over-abstracted spatial features. Therefore, D is set to 2 in this study for stable training and testing performance.

B. Ablation Experiments

1) Effectiveness of GLR Branches: To demonstrate the effectiveness of the proposed GLR branches, ablation experiments are conducted on four HSI datasets. These experiments involve six different ablation approaches, ranging from single to multiple feature extraction, to verify the effectiveness of GLR-G, GLR-L, and GLR-R branches. G and L represent the models that input a patch through the GLR-G or GLR-L branch for feature extraction, respectively. These are the networks composed of separate operator with single feature extraction capability. GR and LR represent the models that input a patch through the GLR-G or GLR-L branch with residual structure for feature extraction, respectively. They are employed to verify whether the feature ability is improved after the joining of GLR-R. GL represents the model that inputs a patch through GLR-G and GLR-L branches for feature extraction. It is employed to analyze the fusion effectiveness of different operators. GLR represents the GLRFNet model with a complete feature extraction process. The ablation results are obtained under the condition with 20 training samples per class.

The results of ablation experiments are presented in Table VI. It is evident that the classification performance improves as the combination of GLR-G, GLR-L, and GLR-R becomes more comprehensive. This improvement can be attributed to the more powerful extraction and the enhanced feature quality. First, when comparing the individual feature extraction capabilities of G and L, it is observed that their performance is approximate for most datasets. The combined GL, which integrates the global and local capabilities, consistently outperforms the single G or L. The fusion of attention and convolution mechanisms is beneficial for information complementarity and feature recognition. The OA and Kappa are raised significantly on the NYCSU dataset, and the OA of GL is 13.8% and 14.65% higher than that of G and L, respectively. It exhibits more complexity and confusion of tree categories in this dataset, with similar features and intricate distribution. Second, when comparing LR and GR with L and G, it can be seen that the introduction of GLR-R improves the feature capability of GLR-G and GLR-L. The OA of GR is 1.3%, 6.96%, 1.13%, and 0.98% higher than that of G on four datasets, respectively. The OA of LR is 1.23%, 14.14%, 1.23%, and 1.22 higher than that of L on four datasets, respectively. The residual connection helps reduce the difficulty of parameter training and improve the feature transfering, for both global and local information extraction. Finally, the exceptional feature extraction capability of GLR is demonstrated by comparing with the other five ablation models, which validates the rationality of integration for GLR-G, GLR-L, and GLR-R branches.

2) Effectiveness of IBF Module: To validate the superiority of IBF over the traditional methods for early fusion, three ablation approaches are established: addition, concatenation, and IBF. Specifically, when dealing with GR_{ij} and LR_{ij}, the addition approach combines two feature vectors by performing $O_{ij} = \alpha(GR_{ij}) + \beta(LR_{ij})$, where α and β are the learnable parameters. The concatenation approach merges GR_{ij} and LR_{ij} along the channel dimension, performing $O_{ij} = [\alpha(GR_{ij}), \beta(LR_{ij})]$. On the other hand, IBF completes the nonlinear fusion process of $\alpha(GR_{ij})$ and $\beta(LR_{ij})$ following the optimized bottleneck operations, as proposed in GLRFNet.

The classification results of addition, concatenation, and IBF module on four datasets are presented in Table VII. As expected, IBF outperforms addition and concatenation for

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TABLE VI
ABLATION STUDY RESULTS OF GLR BRANCHES ON FOUR DATASETS

Dataset	Indicator	G	L	GR	LR	GL	GLR
SYCSU	OA (%)	89.63±1.32	88.80±1.45	90.93±1.27	90.03±1.34	91.55±1.18	**93.43±1.05**
	Kappa	0.881±0.015	0.870±0.018	0.895±0.014	0.885±0.016	0.902±0.012	**0.922±0.011**
NYCSU	OA (%)	75.80±2.34	74.95±2.48	82.76±1.94	89.09±1.56	89.60±1.37	**90.87±1.17**
	Kappa	0.710±0.021	0.681±0.025	0.791±0.019	0.864±0.015	0.871±0.013	**0.887±0.012**
YRE	OA (%)	82.15±1.87	87.72±1.41	83.28±2.06	88.95±1.62	89.81±1.45	**92.14±1.21**
	Kappa	0.794±0.017	0.858±0.013	0.807±0.020	0.872±0.016	0.882±0.014	**0.909±0.013**
YC	OA (%)	93.37±1.12	92.61±1.25	94.35±1.08	93.83±1.14	95.03±0.96	**96.65±0.93**
	Kappa	0.921±0.012	0.912±0.014	0.932±0.011	0.926±0.013	0.940±0.009	**0.960±0.008**

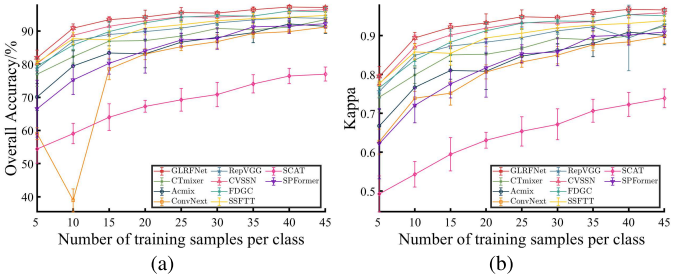
TABLE VII
ABLATION STUDY RESULTS OF IBF MODULE ON FOUR DATASETS

Dataset	Indicator	Addition	Concatenation	IBF
SYCSU	OA (%)	90.17±1.28	89.42±1.36	**93.43±1.05**
	Kappa	0.887±0.014	0.878±0.015	**0.922±0.011**
NYCSU	OA (%)	88.38±1.49	87.59±1.57	**90.87±1.17**
	Kappa	0.857±0.016	0.847±0.017	**0.887±0.012**
YRE	OA (%)	88.95±1.41	89.20±1.38	**92.14±1.21**
	Kappa	0.872±0.013	0.875±0.013	**0.909±0.013**
YC	OA (%)	94.00±1.11	94.27±1.09	**96.65±0.93**
	Kappa	0.928±0.010	0.931±0.010	**0.960±0.008**

HSI classification tasks. On average, IBF achieves an OA that is 3.64%, 2.89%, 3.07%, and 2.52% higher than that of the other two approaches on four datasets, respectively. This confirms that IBF significantly enhances the effectiveness of early fusion and further improves the classification performance. Moreover, the accuracy of addition and concatenation is approximate, which belong to relatively direct fusion methods rather than deep nonlinear fusion methods. The addition is a little more effective than concatenation on the SYCSU and NYCSU datasets, but a little less effective than concatenation on the YRE and YC datasets. This discrepancy may be attributed to the variations in resolution and ground objects present in different datasets. SYCSU and NYCSU datasets have higher resolutions and consist of similar tree categories, while YRE and YC datasets have lower resolutions and encompass a wider range of ground targets.

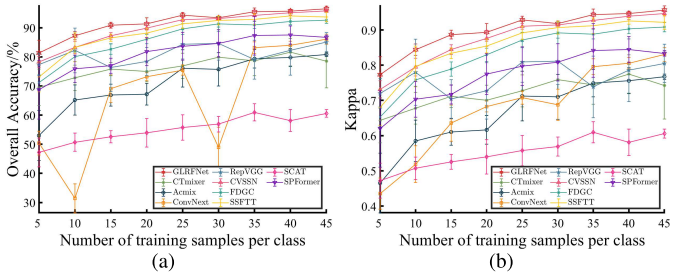
C. Comparison Experiments

1) Introduction of Comparison Methods: Nine state-of-the-art methods are conducted as comparison experiments, including CTmixer [33], Acmix [34], ConvNext [49], RepVGG [50], SCAT [51], SPFormer [52], SSFTT [53], FDGC [54], and CVSSN [55]. CTmixer and Acmix are both representative of methods with multiple operators, which are closely related to the proposed GLRFNet. ConvNext and CVSSN are deep learning networks based on convolution with advanced designs. ConvNext resembles Swin-Transformer in terms of macroscopic and microscopic details, while CVSSN incorporates a self-similarity mechanism to enhance the relationship among pixels. RepVGG and FDGC are multibranch networks. RepVGG employs different kernels, and FDGC integrates CNN and graph convolution branches to jointly extract spatial–spectral features. SSFTT, SCAT, and SPFormer demonstrate the application of attention mechanism in classification tasks. SPFormer is a lightweight Transformer designed for few-shot HSI classification, leveraging self-supervised dimensionality reduction and parameter-free modules.



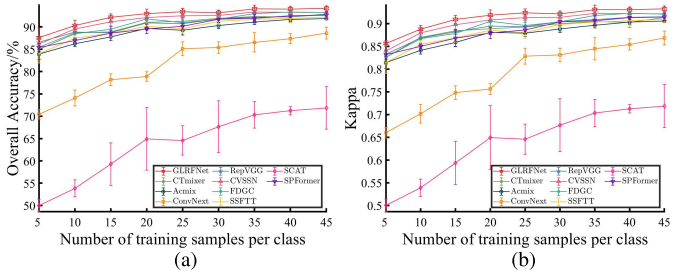
(a) (b)

Fig. 6. Classification accuracy versus different number of training samples per class on the SYCSU dataset in terms of (a) OA (%) and (b) Kappa.



(a) (b)

Fig. 7. Classification accuracy versus different number of training samples per class on the NYCSU dataset in terms of (a) OA (%) and (b) Kappa.



(a) (b)

Fig. 8. Classification accuracy versus different number of training samples per class on the YRE dataset in terms of (a) OA (%) and (b) Kappa.

2) Classification Results: Figs. 6–9 illustrate the overall performance of each comparison method on four datasets with different numbers of training samples. As shown in figures, the performance of all methods improves as the number of training samples increases, but it reaches a saturation point once sufficient samples are available. Specifically, SCAT consistently

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TABLE VIII

CLASSIFICATION ACCURACY (%) AND KAPPA USING SCAT, CONVNEXT, REPVGG, ACMIX, CTMIXER, SPFORMER, SSFTT, FDGC, CVSSN, AND GLRFNET FOR THE SYCSU DATASET WITH 20 TRAINING SAMPLES PER CLASS

Class	SCAT	ConvNext	RepVGG	Acmix	CTmixer	SPFormer	SSFTT	FDGC	CVSSN	GLRFNet
C1	75.60±3.41	89.52±1.36	88.64±1.25	87.07±1.12	**97.45±1.12**	85.97±5.57	89.36±5.17	93.44±3.43	90.73±4.56	88.87±1.23
C2	63.54±4.85	69.28±2.85	86.02±2.41	74.73±3.12	79.17±3.34	74.29±5.87	95.38±3.75	94.65±2.84	**96.47±2.41**	95.57±0.98
C3	87.03±1.76	99.65±0.11	97.66±0.89	90.40±2.06	**99.86±0.08**	92.27±8.20	96.25±4.21	98.69±1.33	97.44±2.56	99.66±0.24
C4	95.79±0.42	98.78±0.78	96.04±1.15	97.05±1.34	96.83±1.45	90.39±3.39	97.21±5.59	97.21±5.59	97.21±5.59	**100.00±0.00**
C5	47.52±5.23	60.66±4.21	83.98±3.19	70.78±3.67	75.21±3.65	68.17±12.78	77.72±9.44	79.30±9.74	83.75±8.07	**89.66±1.48**
C6	76.47±3.95	94.85±1.25	**100.00±0.00**	99.94±0.05	99.55±0.18	99.37±0.64	**100.00±0.00**	99.98±0.04	**100.00±0.00**	**100.00±0.00**
C7	86.24±2.15	90.62±2.18	99.09±0.24	97.55±1.07	99.15±0.24	95.70±3.25	**100.00±0.00**	99.94±0.13	**100.00±0.00**	**100.00±0.00**
C8	58.89±4.63	93.22±1.89	91.36±2.01	82.07±2.56	88.85±2.36	79.79±8.65	95.52±3.41	95.82±5.19	96.06±3.99	**99.99±0.01**
C9	69.84±3.72	75.86±2.54	83.89±2.32	75.53±2.68	78.26±3.42	74.08±8.16	83.99±6.79	85.03±8.81	90.60±7.53	**96.71±0.87**
C10	76.49±2.98	85.95±2.08	97.68±0.86	95.61±1.21	92.50±1.89	94.15±5.64	96.95±3.46	98.80±1.69	99.17±0.62	**99.56±0.34**
C11	81.62±2.14	96.97±1.02	96.97±0.97	98.23±0.64	96.04±1.12	91.82±9.83	92.65±3.80	99.56±0.58	98.78±1.65	**100.00±0.00**
C12	43.19±5.38	71.23±3.08	73.60±2.42	74.16±2.95	**82.38±2.98**	60.68±3.47	75.23±6.63	80.66±2.79	82.23±1.32	79.31±3.12
C13	77.73±2.36	91.00±1.95	98.19±0.86	97.24±0.78	98.29±0.76	98.73±0.99	95.35±5.47	99.23±1.24	98.71±1.58	**99.76±0.18**
C14	91.83±1.24	91.82±1.78	99.25±0.34	97.11±1.25	96.51±1.29	93.43±10.29	98.80±1.64	**100.00±0.00**	99.93±0.15	**100.00±0.00**
OA	64.01±4.18	78.55±3.25	88.95±2.05	83.39±2.14	86.95±2.12	80.28±3.32	87.31±2.74	89.83±3.03	91.42±2.59	**93.43±1.05**
AA	73.70±4.09	86.39±2.27	92.31±1.65	88.39±2.17	91.43±2.09	85.63±5.66	92.46±3.45	94.45±2.70	95.08±2.42	**96.36±0.98**
Kappa	0.595±0.047	0.752±0.035	0.873±0.021	0.811±0.024	0.851±0.030	0.776±0.036	0.855±0.030	0.883±0.034	0.901±0.029	**0.922±0.011**

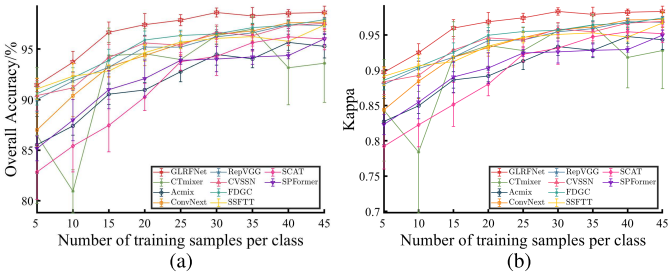


Fig. 9. Classification accuracy versus different number of training samples per class on the YC dataset in terms of (a) OA (%) and (b) Kappa.

exhibits relatively poor performance due to its limited ability to extract local features, which constrains its recognition capability. SPFormer incorporates lightweight Transformer features, optimizing performance for small-sample tasks, but its potential remains underutilized in large-sample scenarios. In contrast, convolution-based methods such as ConvNext and RepVGG demonstrate local feature extraction abilities, achieving performance gains when ample samples are available. However, they exhibit high sensitivity to sample distribution, leading to greater fluctuations under limited sample conditions. CVSSN enhances neighborhood relationship modeling via self-similarity mechanisms, showing competitive performance but still falling short compared to GLRFNet. SSFTT and FDGC perform well in scenarios with limited samples, but due to the design complexity, their performance growth plateaus under abundant sample conditions.

CTmixer and Acmix, as methods with a structure of different operators, show stable performance with fewer training samples. However, as the number of training samples increases, their advantages gradually diminish, and eventually they are unable to outperform most methods. This is because CTmixer and Acmix stack the features extracted by different operators without an effective fusion process. By comparing with all the methods, it is evident that GLRFNet consistently exhibits the best classification performance across all datasets and training sample sizes. It validates the effectiveness of improvements made in GLRFNet, including information projection with multiple kernels, GLR feature extraction, and optimized inverted bottleneck feature fusion. The SYCSU and NYCSU datasets have more confused spectral features of tree species and more refined spatial details of high resolution compared to the YRE and YC datasets. Dense and complex distributions of tree species place higher demands on the discriminative capabilities of classification methods. Accordingly, there is a significant difference in accuracy between various methods on the SYCSU and NYCSU datasets, while on the YRE and YC datasets, some methods have similar accuracy. However, GLRFNet surpasses all the comparison methods on four datasets, which demonstrates the generalizability of GLRFNet and its robust performance when faced with complex classification tasks.

3) Performance Under Limited Samples: Tables VIII–XI present the CA, OA, AA, and Kappa for each method's performance on four datasets with 20 training samples per class. In general, GLRFNet achieves the best performance in terms of OA and Kappa coefficient, under the condition with limited samples. Although GLRFNet does not obtain the highest AA on the NYCSU and YRE datasets, it achieves the highest CA on the main categories. In the SYCSU dataset, C1 (*Phoenix sylvestris*), C5 (*Terminalia mantaly*), and C12 (*Camphora officinarum*) are more challenging to classify, due to category similarity between C1 and C8 (*Cocos nucifera*) and irregular distribution of C5 and C12. GLRFNet achieves the best classification results and shows significant differences from other methods on C5 and C9 (*Delonix regia*). Additionally, GLRFNet achieves 100% accuracy on C4 (*Mangifera indica*), C6 (*Araucaria cunninghamii*), C7 (*Catalpa*), C11 ((*Plumeria*), and C14 ((*Bombax ceiba*). In the NYCSU dataset, C1 (*Microcarpa*) has the largest sample size and complex distribution, and GLRFNet outperforms the second-best by 4.42%. C5 (*Acacia auriculiformis*) is another species that is difficult to distinguish, and GLRFNet exceeds the second-best by 3.74%. Due to intricate distribution and complex situation of different tree species in the NYCSU scene, most methods struggle to extract features and implement classification.

In the YRE dataset, GLRFNet achieves the highest CA on eight categories, and CVSSN and CTmixer have the best CA on five and four categories, respectively. This demonstrates

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TABLE IX

CLASSIFICATION ACCURACY (%) AND KAPPA USING SCAT, CONVNEXT, REPVGG, ACMIX, CTMIXER, SPFORMER, SSFTT, FDGC, CVSSN, AND GLRFNET FOR THE NYCSU DATASET WITH 20 TRAINING SAMPLES PER CLASS

Class	SCAT	ConvNext	RepVGG	Acmix	CTmixer	SPFormer	SSFTT	FDGC	CVSSN	GLRFNet
C1	44.18±4.12	55.17±3.54	83.54±1.68	55.86±2.78	61.76±3.25	72.99±3.25	82.87±3.73	74.53±4.60	77.20±2.45	**87.96±1.89**
C2	79.83±2.95	83.62±2.73	77.30±3.12	91.03±1.47	83.25±2.14	93.50±2.10	95.75±5.56	94.45±3.55	95.40±1.86	**97.80±0.87**
C3	63.25±3.89	81.33±2.11	69.02±3.74	62.27±2.64	84.59±1.97	91.05±2.37	89.56±11.85	88.23±4.55	**94.59±1.72**	88.74±1.76
C4	61.43±3.28	86.88±1.56	96.31±0.74	84.67±1.88	94.91±0.96	95.59±1.88	96.05±2.77	93.11±2.25	**99.49±0.31**	97.61±0.92
C5	48.78±4.52	69.76±3.21	50.01±4.13	73.40±2.47	80.07±2.73	71.66±3.45	83.19±4.14	84.29±5.05	80.14±3.12	**88.03±2.02**
C6	71.89±2.59	80.14±1.91	55.97±3.76	68.74±2.68	83.71±1.86	75.55±3.12	92.78±6.13	85.52±4.25	**95.31±1.45**	87.88±1.87
C7	67.67±3.12	63.83±3.52	57.50±3.34	76.11±2.41	87.38±1.59	63.03±4.25	81.98±9.53	87.69±5.48	**95.81±1.27**	89.73±1.58
C8	48.58±4.45	70.92±2.76	51.97±4.12	53.71±3.63	82.66±2.07	68.64±3.56	82.78±5.95	81.92±2.24	**99.69±0.15**	89.47±1.89
C9	37.92±4.87	75.92±2.34	55.23±3.89	71.68±2.83	**97.73±0.54**	69.15±2.67	82.24±17.11	82.16±5.21	87.73±2.08	95.87±0.92
C10	39.70±4.13	86.33±1.78	82.52±1.97	72.49±2.16	89.70±1.48	85.54±1.98	91.48±4.68	96.82±2.62	96.99±0.78	**97.98±0.76**
C11	63.79±2.68	82.85±1.93	83.52±1.74	77.54±2.12	72.59±2.89	71.10±3.02	90.48±6.37	84.94±5.62	91.25±2.65	**91.65±1.32**
OA	52.56±3.41	69.05±3.34	76.34±2.12	66.87±2.79	75.80±2.56	76.96±2.45	86.56±1.25	82.64±1.75	87.21±3.20	**90.87±1.17**
AA	57.00±5.81	76.07±4.63	69.35±6.24	71.59±5.47	83.49±4.28	77.98±4.91	88.11±3.76	86.70±3.98	**92.15±2.74**	92.07±2.81
Kappa	0.525±0.032	0.636±0.032	0.704±0.022	0.610±0.031	0.712±0.030	0.717±0.029	0.834±0.015	0.789±0.020	0.844±0.030	**0.887±0.012**

TABLE X

CLASSIFICATION ACCURACY (%) AND KAPPA USING SCAT, CONVNEXT, REPVGG, ACMIX, CTMIXER, SPFORMER, SSFTT, FDGC, CVSSN, AND GLRFNET FOR THE YRE DATASET WITH 20 TRAINING SAMPLES PER CLASS

Class	SCAT	ConvNext	RepVGG	Acmix	CTmixer	SPFormer	SSFTT	FDGC	CVSSN	GLRFNet
C1	34.93±7.05	48.82±6.28	69.72±5.43	78.84±4.15	**82.96±4.12**	78.87±4.90	79.14±7.49	66.71±14.20	76.84±10.21	79.55±3.88
C2	89.62±3.11	89.42±3.63	95.48±2.12	92.86±1.83	**99.96±0.18**	86.63±7.43	88.79±5.08	99.34±0.49	97.58±1.40	95.83±1.82
C3	74.91±4.87	45.52±7.56	77.46±3.65	74.56±4.67	70.23±5.74	79.75±2.22	77.67±5.75	80.75±2.66	81.07±6.36	**83.92±4.15**
C4	68.77±4.63	92.71±2.84	96.65±1.99	96.88±2.29	89.10±3.82	93.48±3.25	92.08±3.14	94.92±4.10	96.97±1.77	**97.26±1.25**
C5	68.48±5.21	97.52±1.52	99.07±0.92	98.56±0.84	99.33±1.05	96.74±3.00	97.62±1.72	98.36±1.91	98.21±1.77	**99.44±0.76**
C6	65.91±4.97	80.57±3.23	71.99±4.97	82.54±3.77	83.27±4.29	70.99±14.69	81.36±6.31	**86.38±1.67**	81.61±8.19	76.87±4.31
C7	51.17±6.89	60.84±4.95	69.27±5.49	54.35±6.72	69.16±5.92	68.02±3.58	63.97±7.89	62.93±4.28	67.27±2.66	**71.04±4.94**
C8	31.11±7.48	70.87±5.81	93.96±1.62	92.53±1.88	91.33±2.24	**94.96±3.92**	90.91±6.50	92.14±2.62	92.76±4.97	86.96±2.95
C9	30.67±7.82	57.21±7.95	88.81±3.42	86.59±4.13	89.46±4.67	90.06±5.85	89.49±5.65	84.25±7.23	**91.31±3.16**	89.78±2.38
C10	43.98±7.04	53.53±6.15	61.19±5.48	66.18±5.52	82.19±4.14	84.45±3.83	84.35±4.93	81.07±8.64	**89.20±4.13**	77.00±3.57
C11	28.45±7.78	72.50±5.04	89.23±3.16	90.65±3.13	86.73±3.83	96.35±2.15	95.98±3.59	95.93±2.78	92.78±3.71	**97.43±1.09**
C12	40.25±6.65	65.72±5.01	93.13±1.51	91.07±1.92	86.52±3.12	90.98±4.75	90.92±5.42	85.73±4.30	87.32±3.64	**96.96±1.01**
C13	35.69±6.15	55.53±5.73	79.95±3.59	76.20±4.13	88.29±4.09	86.22±5.35	84.53±4.01	73.78±15.81	86.61±5.18	**90.90±2.36**
C14	36.73±7.02	46.28±6.73	73.59±4.81	60.15±5.21	**90.11±4.56**	73.72±4.03	79.62±5.54	75.95±13.76	78.35±4.83	75.25±4.45
C15	49.58±6.51	50.06±6.12	47.54±6.98	55.54±5.17	**95.02±2.73**	57.18±6.41	63.79±8.19	81.10±6.58	74.06±7.31	73.39±4.89
C16	60.09±5.16	68.81±5.62	68.54±4.63	75.58±3.89	74.92±4.58	82.62±10.45	79.23±5.32	89.30±5.89	**90.61±4.57**	86.27±2.46
C17	73.61±5.29	78.17±4.29	89.97±2.14	88.90±4.16	92.07±2.45	**97.79±1.14**	96.00±3.48	95.61±2.07	95.89±4.22	93.10±2.19
C18	20.54±7.83	36.28±5.73	44.94±5.92	39.17±7.12	51.49±6.13	53.13±4.51	52.26±8.41	54.33±11.35	**62.84±9.04**	52.94±6.25
C19	56.33±5.24	77.58±3.92	69.22±5.52	70.72±4.65	65.76±5.24	73.15±6.64	75.90±4.12	84.28±5.34	**85.44±5.02**	67.44±4.18
C20	59.64±5.03	86.93±3.15	94.40±2.81	95.20±1.92	68.00±4.10	93.82±9.87	90.18±10.13	88.00±12.89	90.55±8.71	**100.00±0.00**
OA	59.28±5.62	78.17±4.47	88.70±2.18	87.78±3.34	88.84±3.08	88.57±1.62	88.61±0.85	89.48±1.82	91.16±1.01	**92.14±1.21**
AA	51.02±3.12	66.74±2.85	78.71±2.43	78.35±2.56	82.79±2.21	82.45±2.33	82.69±2.18	83.54±2.09	**85.86±1.97**	84.57±1.05
Kappa	0.593±0.039	0.748±0.036	0.882±0.022	0.859±0.027	0.884±0.025	0.868±0.018	0.868±0.010	0.878±0.021	0.898±0.012	**0.909±0.013**

TABLE XI

CLASSIFICATION ACCURACY (%) AND KAPPA USING SCAT, CONVNEXT, REPVGG, ACMIX, CTMIXER, SPFORMER, SSFTT, FDGC, CVSSN, AND GLRFNET FOR THE YC DATASET WITH 20 TRAINING SAMPLES PER CLASS

Class	SCAT	ConvNext	RepVGG	Acmix	CTmixer	SPFormer	SSFTT	FDGC	CVSSN	GLRFNet
C1	98.08±1.14	99.77±0.25	96.43±1.87	97.45±2.23	93.54±3.45	96.01±3.84	97.10±1.68	96.00±2.45	**100.00±0.00**	**100.00±0.00**
C2	99.49±0.52	**100.00±0.00**	99.86±0.38	99.66±0.45	99.42±0.73	98.49±1.19	99.41±0.57	94.86±1.32	99.39±0.50	**100.00±0.00**
C3	86.89±5.02	96.55±3.28	91.27±4.11	88.18±6.34	89.64±5.21	96.22±2.78	98.22±1.13	**100.00±0.00**	99.78±0.44	90.91±4.14
C4	68.14±7.52	90.68±4.63	96.24±2.15	97.14±2.19	97.14±1.99	96.64±3.14	91.33±6.06	97.35±1.88	98.41±3.19	**100.00±0.00**
C5	80.00±7.89	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C6	90.79±3.25	97.11±2.21	94.94±3.17	91.57±4.98	96.87±2.72	85.40±17.69	98.10±1.56	**100.00±0.00**	98.41±2.46	87.95±4.11
C7	64.04±7.19	92.48±4.52	84.31±6.29	79.17±5.36	96.42±3.18	83.74±11.23	89.70±7.57	96.97±3.01	91.41±3.78	**99.54±1.29**
C8	70.29±6.31	91.40±3.71	93.07±2.83	91.05±3.56	**97.81±1.99**	90.38±4.72	88.94±3.88	87.50±4.12	87.12±6.11	97.37±2.24
C9	75.44±4.96	83.32±5.79	82.74±6.11	73.49±6.84	96.90±3.41	72.36±7.42	76.23±4.41	**97.04±1.45**	75.17±9.02	84.50±6.03
C10	75.79±6.03	95.47±2.15	99.30±0.89	97.79±1.63	98.60±1.55	99.87±0.26	98.42±2.11	**100.00±0.00**	98.95±2.11	**100.00±0.00**
C11	93.91±3.31	95.48±3.51	96.68±2.59	89.59±4.28	**98.80±1.27**	92.99±2.21	94.42±2.96	92.89±2.67	92.99±2.47	95.39±2.97
C12	80.44±7.65	93.59±3.17	**100.00±0.00**	97.95±2.02	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C13	87.91±5.38	95.71±3.19	99.74±0.48	99.48±0.72	63.25±7.89	**100.00±0.00**	99.55±0.90	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C14	65.12±7.39	58.83±8.31	86.06±5.52	78.06±6.72	88.44±5.51	87.41±14.10	90.65±7.11	66.47±5.29	**92.94±7.44**	91.94±3.76
C15	59.81±6.82	78.72±5.64	70.60±6.12	58.21±7.23	**90.85±4.61**	60.65±15.50	72.62±13.14	85.98±2.84	72.24±11.25	88.89±4.34
C16	76.96±4.23	80.00±6.34	78.43±5.27	61.08±5.88	75.00±7.21	74.89±4.16	76.74±4.35	84.78±3.19	84.46±4.89	**92.16±2.87**
C17	80.00±7.73	94.82±3.85	93.88±3.17	92.94±3.12	**100.00±0.00**	90.46±4.61	92.31±5.33	95.38±1.73	95.69±4.06	**100.00±0.00**
C18	81.33±6.51	97.71±2.42	**100.00±0.00**	**100.00±0.00**	99.43±0.58	97.33±5.33	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C19	95.56±2.61	98.95±1.91	**100.00±0.00**	93.68±2.95	**100.00±0.00**	92.22±15.56	91.11±10.89	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
C20	80.54±6.37	96.52±2.13	95.30±2.99	97.58±2.74	94.39±3.29	97.68±2.08	96.96±4.50	98.21±0.94	95.54±1.96	**99.24±1.88**
OA	87.44±4.59	93.22±3.15	93.21±3.71	90.53±4.21	94.35±3.32	90.99±1.89	92.81±0.63	93.88±1.56	94.17±1.28	**96.65±0.93**
AA	80.53±2.85	91.86±1.37	92.94±1.26	89.20±1.74	93.83±1.14	90.64±1.62	92.59±1.35	94.67±1.02	94.13±1.08	**96.39±1.03**
Kappa	0.851±0.037	0.918±0.026	0.919±0.033	0.886±0.035	0.919±0.029	0.890±0.023	0.912±0.007	0.946±0.021	0.929±0.016	**0.960±0.008**

that methods incorporating diverse operators and hierarchical designs are more effective for feature extraction and category recognition. In the YC dataset, GLRFNet achieves 100% accuracy on categories of C1 (Sea), C2 (Offshore water), C4 (Pond), C5 (Marsh), C10 (Estuarine water), C12 (Woodland), C13 (Bare land), C17 (Suaeda), C18 (Mariculyure), and C19 (Ditch). In addition, other comparison methods also achieve 100% accuracy on some of these categories. YC

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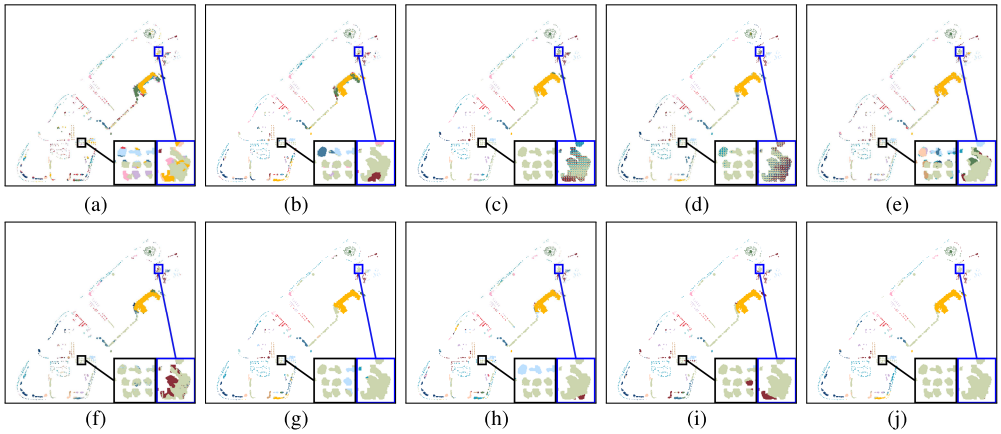


Fig. 10. Classification maps for the SYCSU dataset by (a) SCAT (64.01%), (b) ConvNext (78.55%), (c) RepVGG (88.95%), (d) Acmix (83.39%), (e) CTmixer (86.95%), (f) SPFormer (80.28%), (g) SSFTT (87.31%), (h) FDGC (89.83%), (i) CVSSN (91.42%), and (j) GLRFNet (93.43%) when the number of training samples is 20 per class.

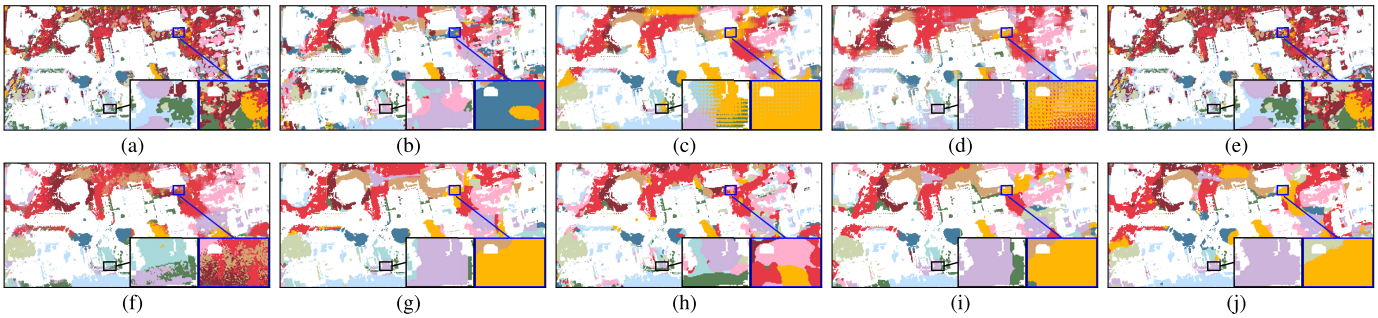


Fig. 11. Classification maps for the NYCSU dataset by (a) SCAT (52.56%), (b) ConvNext (69.05%), (c) RepVGG (76.34%), (d) Acmix (66.87%), (e) CTmixer (75.80%), (f) SPFormer (76.96%), (g) SSFTT (86.56%), (h) FDGC (82.64%), (i) CVSSN (87.21%), and (j) GLRFNet (90.87%) when the number of training samples is 20 per class.

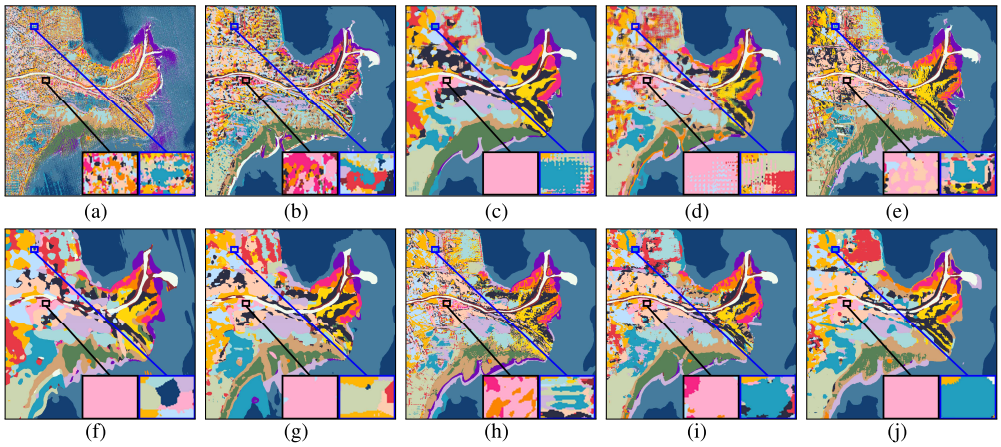


Fig. 12. Classification maps for the YRE dataset by (a) SCAT (59.28%), (b) ConvNext (78.17%), (c) RepVGG (88.70%), (d) Acmix (87.78%), (e) CTmixer (88.84%), (f) SPFormer (88.57%), (g) SSFTT (88.61%), (h) FDGC (89.48%), (i) CVSSN (91.16%), and (j) GLRFNet (92.14%) when the number of training samples is 20 per class.

dataset can achieve higher accuracy than YRE dataset under limited sample conditions because the spectral characteristics of YC dataset are more distinct between categories. SCAT and ConvNext show satisfactory performance in the YC dataset but perform poorly on other datasets. This indicates the unstable performance of these methods when faced with complex tasks, highlighting the superiority of GLRFNet.

4) Visual Comparison: Figs. 10–13 display the classification maps of comparison methods on four datasets using 20 training samples per class. CTmixer and Acmix, despite achieving acceptable classification accuracy, struggle with sparsely distributed land-cover classes and tend to produce chaotic regions, such as C8 (Aquaculture) and C17 (Suaeda) in the YC dataset. RepVGG exhibits obvious noise and

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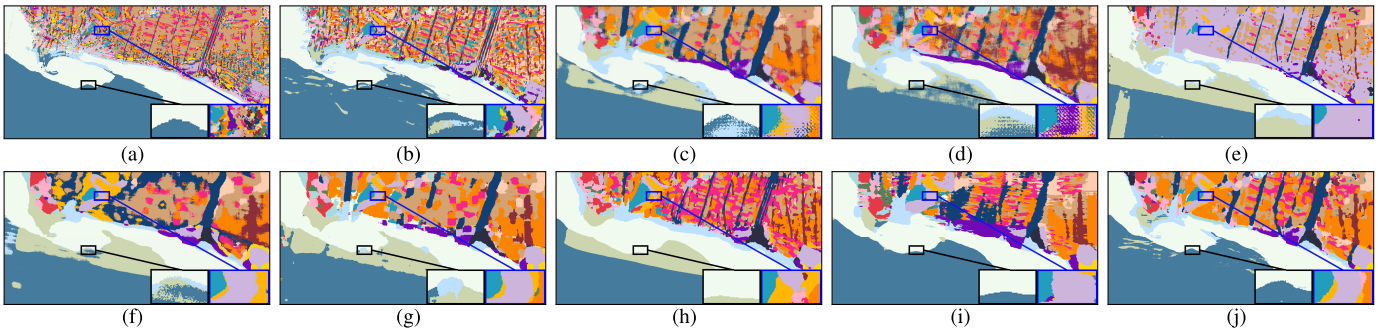


Fig. 13. Classification maps for the YC dataset by (a) SCAT (87.44%), (b) ConvNext (93.22%), (c) RepVGG (93.21%), (d) Acmix (90.53%), (e) CTmixer (94.35%), (f) SPFormer (90.88%), (g) SSFTT (92.81%), (h) FDGC (93.88%), (i) CVSSN (94.17%), and (j) GLRFNet (96.65%) when the number of training samples is 20 per class.

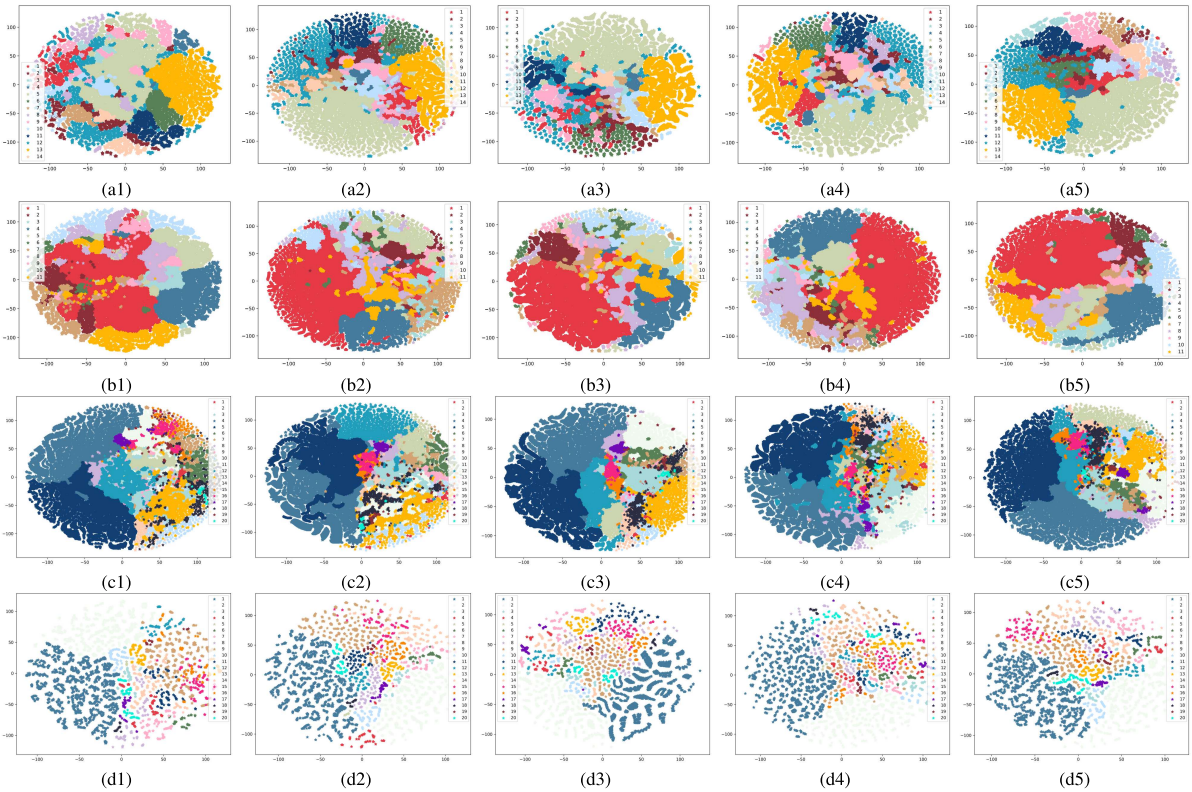


Fig. 14. Feature distribution visualizations for different methods on the four datasets using t-SNE. From top to bottom, (a)–(d) correspond to SYCSU, NYSCU, YRE, and YC datasets, respectively. From left to right, 1–5 show the results of SPFormer, SSFTT, FDGC, CVSSN, and GLRFNet methods, respectively.

ambiguous boundaries, and FDGC exhibits a mix of fragmentation in the classification results since paying too much attention to the neighborhood may ignore the global structure. SCAT produces discrete classification maps with visible stripes in the YRE dataset, further weakening the spatial consistency. ConvNext and CVSSN reduce the salt-and-pepper effect but show limitations in distinguishing fine-grained tree species in the SYCSU and NYCSU datasets. SSFTT and SPFormer improve spatial consistency to some extent but still face issues of blurred boundaries and local misclassification. In comparison, GLRFNet excels in both classification accuracy and spatial consistency. Its results demonstrate coherent and complete class regions with clear boundaries. GLRFNet achieves high-precision recognition for both densely and sparsely distributed land-cover classes, highlighting its significant advantages in feature extraction and spatial modeling.

In addition, t-SNE is employed to project the output features onto a 2-D plane, with samples of different classes represented by points of distinct colors. As shown in Fig. 14, the visualization facilitates an intuitive comparison of feature separability between the proposed and comparison methods. Four comparison methods with relatively high classification accuracy are selected for feature analysis. Specifically, for C12 (*C. officinarum*) in the SYCSU dataset and C10 (*C. officinarum*) in the NYCSU dataset, the features of SPFormer, SSFTT, FDGC, and CVSSN are scattered into multiple subclusters, which is easy to be confused with other categories. However, it achieves a more compact aggregation under GLRFNet,

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TABLE XII
TRAINING AND INFERENCE TIME (S) OF DIFFERENT METHODS ON FOUR DATASETS

Dataset	Time	SCAT	ConvNext	RepVGG	Acmix	CTmixer	SPFormer	SSFTT	FDGC	CVSSN	GLRFNet
SYCSU	Train(s)	538.89	406.48	336.29	394.69	669.00	684.24	685.80	529.42	504.58	533.00
	Inference(s)	426.41	445.77	416.23	448.52	342.05	691.18	506.67	407.66	421.61	422.57
NYCSU	Train(s)	407.14	281.10	249.03	285.70	504.50	535.89	485.35	372.27	375.17	419.51
	Inference(s)	2371.99	2228.65	2105.19	2386.27	2068.46	4421.97	3020.63	2017.92	2389.78	2378.63
YRE	Train(s)	232.36	229.37	220.89	204.37	315.62	142.52	224.15	246.77	205.01	231.48
	Inference(s)	1265.07	1325.25	1315.54	1432.13	1557.03	694.98	1219.24	1260.85	1180.94	1284.26
YC	Train(s)	202.48	222.52	191.87	199.17	274.43	153.62	203.06	221.48	183.22	218.96
	Inference(s)	499.73	534.50	504.95	528.37	586.00	285.08	459.67	491.76	455.48	503.46

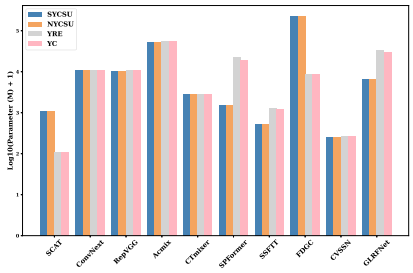


Fig. 15. Number of parameters for comparison methods on the SYCSU, NYCSU, YRE, and YC datasets.

indicating reduced intraclass variation. In the YRE dataset, in addition to C4 (Shallow sea) and C5 (Deep sea) with a large number of samples have good feature aggregation, the other class features of the comparison methods have obvious mixed distribution, especially C13 (Rice) and C18 (Reed). In the YC dataset, C8 (Aquculture) class exhibits relatively dispersed feature distributions in the comparison methods but shows greater feature compactness under GLRFNet. Overall, the comparative analysis of feature separability across the four datasets clearly demonstrates that GLRFNet achieves larger interclass distances and smaller intraclass distances compared to the comparison methods, thereby exhibiting superior feature separability. These results further validate the notable advantages of GLRFNet in HSI classification tasks.

5) Efficiency Analysis: Fig. 15 illustrates the parameter comparison between GLRFNet and other methods on the four datasets. Table XII displays the training and inference times, where the training time is based on a total of 200 epochs using 20 samples per class. These efficiency results are influenced by factors including spectral band number, patch size, category number, and model architecture. Overall, GLRFNet demonstrates a good balance between complexity and performance across different datasets, showing high adaptability and efficiency for various scenes and conditions, particularly on the SYCSU and NYCSU datasets. However, GLRFNet has certain limitations, such as its sensitivity to the number of spectral bands. On the YRE and YC datasets with a larger number of bands, GLRFNet achieves optimal performance but exhibits a higher parameter complexity. CVSSN maintains a relatively small number of parameters on different datasets, which is not greatly affected by spatial resolution and band number. Models that use parameter sharing or dimensionality reduction have reduced parameters as the band number increases, while models that rely on high-dimensional transformations have increased parameters. SPFormer has a longer runtime on the SYCSU and NYCSU datasets, but runs faster on the YRE and YC datasets. In comparison, GLRFNet demonstrates a balanced runtime performance, with both training and inference times being reasonable relative to the other models.

VI. CONCLUSION

In this article, a novel GLRFNet is proposed to improve the HSI classification. It creates a feature pool that consists of shallow features from different receptive fields, enhancing the coupling between operators and expanding the diversity of features. The GLR block is designed to extract features with varying characteristics in parallel, striking a balance between linear and nonlinear processing, while emphasizing the extraction of both local and global features. The IBF block is optimized and enhanced with a series of nonlinear operations, enabling the efficient fusion of features with diverse characteristics. The GLRFNet outperforms other state-of-the-art methods across different sample sizes on four HSI datasets with varying scenes and resolutions. It demonstrates excellent performance with limited labeled samples for training, showing higher classification accuracy and better category boundaries. In future work, novel fusion operators, such as graph convolution-based methods, will be explored to enhance the feature relationship modeling. Lightweight modules and compression techniques will be investigated to reduce the computational complexity, making GLRFNet suitable for resource-constrained environments. Additionally, integrating GLRFNet into a multitask learning framework could enable simultaneous classification and other tasks, enhancing its practicality in complex applications and inspiring further research across diverse HSI scenarios.

REFERENCES

[1] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[2] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[3] S. Zhang, J. Li, Z. Wu, and A. Plaza, "Spatial discontinuity-weighted sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5767–5779, Oct. 2018.

[4] W. He, X. Fu, N. Li, Q. Ren, and S. Jia, "LGCT: Local–global collaborative transformer for fusion of hyperspectral and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537114.

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[5] X. Qiao, L. L. Li, D. Li, Y. L. Gan, and A. Y. Hou, "Application of machine learning in urban greenery land cover extraction," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 1409–1413, Apr. 2018.

[6] Q. Feng, J. Liu, and J. Gong, "UAV remote sensing for urban vegetation mapping using random forest and texture analysis," *Remote Sens.*, vol. 7, no. 1, pp. 1074–1094, Jan. 2015.

[7] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[8] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[9] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[10] B. Du and L. Zhang, "Random-selection-based anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1578–1589, May 2011.

[11] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[12] G. Camps-Valls, L. Gomez-Chova, J. Munoz-Mari, J. Vila-Frances, and J. Calpe-Maravilla, "Composite kernels for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 93–97, Jan. 2006.

[13] M. Fauvel, J. Chanussot, and J. A. Benediktsson, "A spatial–spectral kernel-based approach for the classification of remote-sensing images," *Pattern Recognit.*, vol. 45, no. 1, pp. 381–392, Jan. 2012.

[14] X. Wu, Z. Lai, S. Yu, J. Zhou, Z. Liang, and L. Shen, "Coarse-to-fine low-light image enhancement with light restoration and color refinement," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 1, pp. 591–603, Feb. 2024.

[15] X. Wu, Z. Lai, J. Zhou, X. Hou, W. Pedrycz, and L. Shen, "Light-aware contrastive learning for low-light image enhancement," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 20, no. 9, pp. 1–24, Sep. 2024.

[16] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505114.

[17] N. Li, S. Jiang, J. Xue, S. Ye, and S. Jia, "Texture-aware self-attention model for hyperspectral tree species classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502215.

[18] A. Vaswani, "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 1–11.

[19] A. Jamali, M. Mahdianpari, B. Brisco, D. Mao, B. Salehi, and F. Mohammadimanesh, "3DUNetGSFormer: A deep learning pipeline for complex wetland mapping using generative adversarial networks and Swin transformer," *Ecol. Informat.*, vol. 72, Dec. 2022, Art. no. 101904.

[20] M. B. A. Gibril et al., "Deep convolutional neural networks and Swin transformer-based frameworks for individual date palm tree detection and mapping from large-scale UAV images," *Geocarto Int.*, vol. 37, no. 27, pp. 18569–18599, Dec. 2022.

[21] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[22] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[23] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[24] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[25] W. Teng, X. Wen, N. Wang, and H. Shi, "Tree species classification and mapping based on deep transfer learning with unmanned aerial vehicle high resolution images," *Laser Optoelectron. Prog.*, vol. 56, no. 7, pp. 277–286, 2019.

[26] J. Lin, F. Gao, X. Shi, J. Dong, and Q. Du, "SS-MAE: Spatial–spectral masked autoencoder for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531614.

[27] X. Cao, H. Lin, S. Guo, T. Xiong, and L. Jiao, "Transformer-based masked autoencoder with contrastive loss for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524312.

[28] Y. Zhou, X. Huang, X. Yang, J. Peng, Y. Ban, and N. Jiang, "MSMT-LCL: Multiscale spatial–spectral masked transformer with local contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534316.

[29] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[30] S. Huang, Z. Liu, W. Jin, and Y. Mu, "Superpixel-based multi-scale multi-instance learning for hyperspectral image classification," *Pattern Recognit.*, vol. 149, May 2024, Art. no. 110257.

[31] Y. He, B. Tu, B. Liu, Y. Chen, J. Li, and A. Plaza, "Hybrid multi-scale spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527918.

[32] Y. Ding et al., "Multi-scale receptive fields: Graph attention neural network for hyperspectral image classification," *Expert Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119858.

[33] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[34] X. Pan et al., "On the integration of self-attention and convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 815–825.

[35] P. Tang, M. Zhang, Z. Liu, and R. Song, "Double attention transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[36] B. Zhang, Y. Chen, Y. Rong, S. Xiong, and X. Lu, "MATNet: A combining multi-attention and transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506015.

[37] Z. Yang, Z. Xi, T. Zhang, W. Guo, Z. Zhang, and H.-C. Li, "CMR-CNN: Cross-mixing residual network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8974–8989, 2022.

[38] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.

[39] L. Yang et al., "FusionNet: A convolution–transformer fusion network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 16, p. 4066, Aug. 2022.

[40] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.

[41] R. Xu, X.-M. Dong, W. Li, J. Peng, W. Sun, and Y. Xu, "DBCTNet: Double branch convolution-transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509915.

[42] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.

[43] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

[44] K. Polat and K. Onur Koc, "Detection of skin diseases from dermoscopy image using the combination of convolutional neural network and one-versus-all," *J. Artif. Intell. Syst.*, vol. 2, no. 1, pp. 80–97, 2020.

[45] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[46] Y. Luo, J. Zou, C. Yao, X. Zhao, T. Li, and G. Bai, "HSI-CNN: A novel convolution neural network for hyperspectral image," in *Proc. Int. Conf. Audio, Lang. Image Process. (ICALIP)*, Jul. 2018, pp. 464–469.

[47] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

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[48] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-Graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.
[49] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.
[50] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13733–13742.
[51] N. Li, J. Xue, and S. Jia, "Spectral context-aware transformer for cholangiocarcinoma hyperspectral image segmentation," in *Proc. 5th Int. Conf. Image Graph. Process. (ICIGP)*, 2022, pp. 209–213.
[52] Z. Li, Z. Xue, Q. Xu, L. Zhang, T. Zhu, and M. Zhang, "SPFormer: Self-pooling transformer for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502019.
[53] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[54] Q. Liu, Y. Dong, Y. Zhang, and H. Luo, "A fast dynamic graph convolutional network and CNN parallel network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530215.
[55] M. Li, Y. Liu, G. Xue, Y. Huang, and G. Yang, "Exploring the relationship between center and neighborhoods: Central vector oriented self-similarity network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1979–1993, Apr. 2023.



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